

# US Patent & Trademark Office

## Patent Public Search | Text View

---

|                      |                         |
|----------------------|-------------------------|
| United States Patent | 12385399                |
| Kind Code            | B2                      |
| Date of Patent       | August 12, 2025         |
| Inventor(s)          | Monzon; Byron R. et al. |

---

### Fan blade or vane with improved bird impact capability

---

#### Abstract

A gas turbine engine is provided and includes a first fan blade including a suction surface, a second fan blade comprising a pressure surface and neighboring the first fan blade and a throat region interposed between the suction surface of the first fan blade and the pressure surface of the second fan blade. The throat region includes a passage throat located at a minimum distance between the pressure and suction surfaces. The first and second fan blades are configured such that a pre-compression region is defined in the throat region ahead of the passage throat. Each of the first and second fan blades includes a mean camber line defining a flattened suction surface.

---

**Inventors:** Monzon; Byron R. (Muskego, WI), Power; Bronwyn (New Haven, CT), Joly; Michael M. (Hebron, CT), Elliott; Jason H. (Huntington, IN), Li; Xuetao (South Glastonbury, CT), Miller; Christopher Beaudry (Brooklyn, NY)

**Applicant:** RTX Corporation (Farmington, CT)

**Family ID:** 87567291

**Assignee:** RTX CORPORATION (Farmington, CT)

**Appl. No.:** 18/446078

**Filed:** August 08, 2023

#### Prior Publication Data

|                            |                         |
|----------------------------|-------------------------|
| <b>Document Identifier</b> | <b>Publication Date</b> |
| US 20240052746 A1          | Feb. 15, 2024           |

#### Related U.S. Application Data

us-provisional-application US 63396464 20220809

---

## Publication Classification

**Int. Cl.:** **F01D5/14** (20060101); **F01D5/28** (20060101); **F01D21/04** (20060101); **F02C7/05** (20060101); **F02K3/06** (20060101); **F04D29/32** (20060101)

**U.S. Cl.:**

**CPC** **F01D5/141** (20130101); **F01D5/147** (20130101);

## Field of Classification Search

**CPC:** F01D (5/141); F01D (5/16); F01D (9/041); F01D (5/14); F01D (5/145); F01D (5/142); F01D (5/148); F01D (5/20); F01D (21/045); F01D (5/143); F01D (5/146); F01D (5/147); F01D (5/22); F01D (5/282); F04D (29/324); F04D (29/384); F04D (29/666); F04D (29/544); F05D (2240/301); F05D (2220/36); F05D (2250/70); F05D (2240/304); F05D (2220/32); F05D (2240/30); F05D (2240/303); F05D (2240/306); F05D (2250/713); F05D (2260/961); F05D (2240/302); F05D (2240/305); F05D (2250/71); F05D (2250/72); F05D (2240/124); F05D (2240/122); F05D (2240/121); F05D (2300/603); F05B (2240/301); F02C (7/05); F02K (3/06)

---

## References Cited

### U.S. PATENT DOCUMENTS

| Patent No.   | Issued Date | Patentee Name     | U.S. Cl.  | CPC         |
|--------------|-------------|-------------------|-----------|-------------|
| 3565548      | 12/1970     | Fowler et al.     | N/A       | N/A         |
| 4123196      | 12/1977     | Prince, Jr.       | 415/181   | F04D 21/00  |
| 4431376      | 12/1983     | Lubenstein et al. | N/A       | N/A         |
| 7374403      | 12/2007     | Decker            | 416/223R  | F01D 5/141  |
| 7497664      | 12/2008     | Walter            | 416/DIG.5 | F01D 5/141  |
| 7896619      | 12/2010     | Hill et al.       | N/A       | N/A         |
| 8668456      | 12/2013     | Merriman et al.   | N/A       | N/A         |
| 9051839      | 12/2014     | Senoo             | N/A       | F01D 5/141  |
| 9957804      | 12/2017     | Chouhan           | N/A       | F01D 5/141  |
| 10370976     | 12/2018     | Quach et al.      | N/A       | N/A         |
| 10539032     | 12/2019     | Soni              | N/A       | F01D 9/041  |
| 10718215     | 12/2019     | Warikoo           | N/A       | F01D 9/041  |
| 10859094     | 12/2019     | Hanson            | N/A       | F04D 29/384 |
| 10865806     | 12/2019     | Veitch            | N/A       | F04D 29/668 |
| 11378093     | 12/2021     | Hanson et al.     | N/A       | N/A         |
| 2006/0228206 | 12/2005     | Decker et al.     | N/A       | N/A         |
| 2009/0317227 | 12/2008     | Grover            | 415/208.1 | F01D 5/141  |
| 2011/0202321 | 12/2010     | Lung et al.       | N/A       | N/A         |
| 2014/0017089 | 12/2013     | Ristau et al.     | N/A       | N/A         |
| 2017/0130587 | 12/2016     | Bhaumik           | N/A       | F01D 5/147  |
| 2017/0175556 | 12/2016     | Soni et al.       | N/A       | N/A         |
| 2018/0119706 | 12/2017     | Vogiatzis         | N/A       | N/A         |
| 2020/0102830 | 12/2019     | Gallagher         | N/A       | F02C 3/10   |
| 2020/0141242 | 12/2019     | Nolcheff et al.   | N/A       | N/A         |
| 2020/0158127 | 12/2019     | Nolcheff et al.   | N/A       | N/A         |

|              |         |                  |     |     |
|--------------|---------|------------------|-----|-----|
| 2020/0182074 | 12/2019 | Taniguchi et al. | N/A | N/A |
| 2020/0232330 | 12/2019 | Chuang et al.    | N/A | N/A |
| 2020/0308968 | 12/2019 | Eryilki et al.   | N/A | N/A |
| 2021/0207614 | 12/2020 | Hanson et al.    | N/A | N/A |
| 2024/0052747 | 12/2023 | Monzon et al.    | N/A | N/A |

## FOREIGN PATENT DOCUMENTS

| Patent No. | Application Date | Country | CPC |
|------------|------------------|---------|-----|
| 3070266    | 12/2015          | EP      | N/A |
| 3456920    | 12/2020          | EP      | N/A |
| 3594447    | 12/2020          | EP      | N/A |
| 2474511    | 12/2010          | GB      | N/A |

## OTHER PUBLICATIONS

Partial Search Report issued in European Patent Application No. 23190406.1; Date of Mailing Jan. 15, 2024 (13 pages). cited by applicant

Search Report issued in European Patent Application No. 23190406.1; Date of Mailing May 21, 2024 (14 pages). cited by applicant

Search Report issued in European Patent Application No. 24193423.1; Date of Mailing Jun. 3, 2025 (11 pages). cited by applicant

*Primary Examiner:* Zamora Alvarez; Eric J

*Attorney, Agent or Firm:* CANTOR COLBURN LLP

## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Application No. 63/396,464 filed Aug. 9, 2022, the disclosure of which is incorporated herein by reference in its entirety.

## BACKGROUND

(1) Exemplary embodiments of the present disclosure relate generally to gas turbine engines and, in one embodiment, a fan blade or vane of a gas turbine engine with improved bird impact capability.

(2) In a gas turbine engine, air is compressed in a compressor and compressor air is then mixed with fuel and combusted in a combustor to produce a high-temperature and high-pressure working fluid. This working fluid is directed into a turbine in which the working fluid is expanded to generate power. The generated power drives the rotation of a rotor within the turbine through aerodynamic interactions between the working fluid and turbine blades or airfoils. The rotor can be used to drive rotations of a propeller or to produce electricity in a generator.

(3) In an aircraft, gas turbine engines can be used to generate thrust for the aircraft and are typically supported in nacelles underneath horizontal wings or in nacelles disposed next to vertical wings. In either case, a concern for gas turbine engine health is the prospect of bird impacts on blades or vanes at leading or forward sections of gas turbine engines. This concern is heightened when certain materials are used for those blades or vanes, such as ceramic matrix composites (CMC), which are particularly susceptible to damage from impacts. A further issue is that modifications to the blades or vanes to mitigate damage from impacts cannot have an excessively negative aerodynamic effect on airflows entering the gas turbine engines.

(4) Accordingly, a need exists for a fan blade or vane of a gas turbine engine with improved bird impact capability and limited negative aerodynamic effects.

#### BRIEF DESCRIPTION

- (5) According to an aspect of the disclosure, a gas turbine engine is provided and includes a first fan blade including a suction surface, a second fan blade comprising a pressure surface and neighboring the first fan blade and a throat region interposed between the suction surface of the first fan blade and the pressure surface of the second fan blade. The throat region includes a passage throat located at a minimum distance between the pressure and suction surfaces. The first and second fan blades are configured such that a pre-compression region is defined in the throat region ahead of the passage throat. Each of the first and second fan blades includes a mean camber line defining a flattened suction surface.
- (6) In accordance with additional or alternative embodiments, each of the first and second fan blades includes an airfoil section that is defined on an axisymmetric surface with a symmetry axis of the surface aligned with a gas-path center-line axis.
- (7) In accordance with additional or alternative embodiments, each of the first and second fan blades has a symmetric thickness distribution with a single thickness maxima.
- (8) According to an aspect of the disclosure, a fan blade of a gas turbine engine is provided. The fan blade includes a body having an airfoil shape and exhibiting a range of thickness-over-chord (T/B) values of approximately 0.076 at about 20% span to approximately 0.05 at about 50% span, an average location of max thickness (LMT) of approximately 0.389 at about 0-20% span and an average of leading edge (LE) thickness at 10% chord-over-total chord of approximately 0.044 at 0-20% span.
- (9) In accordance with additional or alternative embodiments, in about a 0%-20% span, the airfoil shape exhibits an average T/B of approximately 0.088 (−5% to +10%), in about a 20%-50% span, the airfoil shape exhibits an average T/B of approximately 0.060 (−5% to +10%), in about a 50%-90% span, the airfoil shape exhibits an average T/B of approximately 0.040 (−5% to +10%) and in about a 90% to 100% span, the airfoil shape exhibits an average T/B of approximately 0.029 (−5% to +10%).
- (10) In accordance with additional or alternative embodiments, in about a 0%-20% span, the airfoil shape exhibits an average LMT of approximately 0.389 (+5% to −10%), in about a 20%-50% span, the airfoil shape exhibits an average LMT of approximately 0.422 (+5% to −10%), in about a 50%-90% span, the airfoil shape exhibits an average LMT of approximately 0.501 (+5% to −10%) and in about a 90% to 100% span, the airfoil shape exhibits an average LMT of approximately 0.556 (+5% to −10%).
- (11) In accordance with additional or alternative embodiments, in about a 0%-20% span, the airfoil shape exhibits an average LE thickness at 10% chord-over-total chord of approximately 0.044 (+0 to 10%), in about a 20%-50% span, the airfoil shape exhibits an average LE thickness at 10% chord-over-total chord of approximately 0.029 (+0 to 10%), in about a 50%-90% span, the airfoil shape exhibits an average LE thickness at 10% chord-over-total chord of approximately 0.019 (+0 to 10%) and in about a 90% to 100% span, the airfoil shape exhibits an average LE thickness at 10% chord-over-total chord of approximately 0.013 (+0 to 10%).
- (12) In accordance with additional or alternative embodiments, the airfoil section exhibits an increasing radial LE angle on a given X-R plane from about a 35% span to about a 75% span.
- (13) In accordance with additional or alternative embodiments, the airfoil section exhibits an inflection in the radial LE angle distribution at about the 75% span above which the radial LE angle decreases relative to the angle at about the 75% span.
- (14) In accordance with additional or alternative embodiments, the airfoil section exhibits an increasing radial LE angle on a given Y-R plane from about a 35% span to about an 80% span.
- (15) In accordance with additional or alternative embodiments, the airfoil section exhibits an inflection in the radial LE angle distribution at about the 80% span above which the radial LE angle decreases relative to the angle at about the 80% span.
- (16) In accordance with additional or alternative embodiments, the airfoil section exhibits a

thickness-to-chord value at a 10% chord location at about 0% LE span of approximately 0.0585, which tapers to approximately 0.0396 at 11% LE span.

(17) In accordance with additional or alternative embodiments, the airfoil section exhibits a chord distribution that has an inflection point between about 50% and about 70% span, the inflection point having a magnitude approximately 1.45-1.55 times a magnitude of the chord at 0% span, and approximately 1-1.1 times a magnitude of the chord at 100% span.

(18) According to an aspect of the disclosure, an airfoil is provided and includes a suction surface exhibiting droop over a first 5% of airfoil chord, after which a suction surface metal-angle distribution is approximately flat along a flat suction surface region, up to approximately -5% of a chord location where an adjacent airfoil covers the airfoil. Following the flat suction surface region, the suction surface metal-angle distribution exhibits a bump between -5% and +5% of an airfoil covered-passage starting position. Following a precompression region, the suction surface metal-angle distribution is approximately linear along a linear section, except at front and end points of this region, where the linear section blends into precompression and trailing-edge locations, respectively.

(19) In accordance with additional or alternative embodiments, an increase in a section maximum-thickness is evidenced on a pressure surface only.

(20) These and other advantages and features will become more apparent from the following description taken in conjunction with the drawings.

---

## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) The following descriptions should not be considered limiting in any way. With reference to the accompanying drawings, like elements are numbered alike:

(2) FIG. 1 is a partial cross-sectional view of a gas turbine engine;

(3) FIG. 2 is a perspective view of an airfoil section of a fan blade in accordance with embodiments;

(4) FIG. 3 is a side view of the airfoil section of FIG. 2 in accordance with embodiments;

(5) FIG. 4 is a schematic illustration of a flow passage formed by neighboring fan blades in accordance with embodiments;

(6) FIG. 5 is a graphical depiction of a range of thickness-over-chord (T/B) values of an airfoil section in accordance with embodiments;

(7) FIGS. 6A and 6B are graphical illustrations of radial leading edge angles of an airfoil section in accordance with embodiments;

(8) FIGS. 7A and 7B are graphical illustrations of radial leading edge angles of an airfoil section in accordance with embodiments;

(9) FIG. 8 is a graphical depiction of a leading edge thickness distribution of an airfoil section in accordance with embodiments; and

(10) FIGS. 9A-9C illustrate a custom tailored airfoil to compensate for the required increases in airfoil thickness for impact-damage resistance while minimizing the negative effects of this thickness on the airfoil aerodynamics in accordance with embodiments.

(11) These and other advantages and features will become more apparent from the following description taken in conjunction with the drawings.

### DETAILED DESCRIPTION

(12) A detailed description of one or more embodiments of the disclosed apparatus and method are presented herein by way of exemplification and not limitation with reference to the Figures.

(13) FIG. 1 schematically illustrates a gas turbine engine 20. The gas turbine engine 20 is disclosed herein as a two-spool turbopump that generally incorporates a fan section 22, a compressor section

**24**, a combustor section **26** and a turbine section **28**. Alternative engines might include other systems or features. The fan section **22** drives air along a bypass flow path B in a bypass duct, while the compressor section **24** drives air along a core flow path C for compression and communication into the combustor section **26** and then expansion through the turbine section **28**. Although depicted as a two-spool turbofan gas turbine engine in the disclosed non-limiting embodiment, it should be understood that the concepts described herein are not limited to use with two-spool turbofans as the teachings may be applied to other types of turbine engines including three-spool architectures.

(14) The exemplary gas turbine engine **20** generally includes a low speed spool **30** and a high speed spool **32** mounted for rotation about an engine central longitudinal axis A relative to an engine static structure **36** via several bearing systems **38**. It should be understood that various bearing systems **38** at various locations may alternatively or additionally be provided, and the location of bearing systems **38** may be varied as appropriate to the application.

(15) The low speed spool **30** generally includes an inner shaft **40** that interconnects a fan **42**, a low pressure compressor **44** and a low pressure turbine **46**. The inner shaft **40** is connected to the fan **42** through a speed change mechanism, which in exemplary gas turbine engine **20** is illustrated as a geared architecture **48** to drive the fan **42** at a lower speed than the low speed spool **30**. The high speed spool **32** includes an outer shaft **50** that interconnects a high pressure compressor **52** and high pressure turbine **54**. A combustor **56** is arranged in the gas turbine engine **20** between the high pressure compressor **52** and the high pressure turbine **54**. The engine static structure **36** is arranged generally between the high pressure turbine **54** and the low pressure turbine **46**. The engine static structure **36** further supports the bearing systems **38** in the turbine section **28**. The inner shaft **40** and the outer shaft **50** are concentric and rotate via bearing systems **38** about the engine central longitudinal axis A which is collinear with their longitudinal axes.

(16) The core airflow is compressed by the low pressure compressor **44** and then the high pressure compressor **52**, is mixed and burned with fuel in the combustor **56** and is then expanded over the high pressure turbine **54** and the low pressure turbine **46**. The high and low pressure turbines **54** and **46** rotationally drive the low speed spool **30** and the high speed spool **32**, respectively, in response to the expansion. It will be appreciated that each of the positions of the fan section **22**, compressor section **24**, combustor section **26**, turbine section **28**, and fan drive gear system **48** may be varied. For example, geared architecture **48** may be located aft of the combustor section **26** or even aft of the turbine section **28**, and the fan section **22** may be positioned forward or aft of the location of geared architecture **48**.

(17) The gas turbine engine **20** in one example is a high-bypass geared aircraft engine. In a further example, the gas turbine engine **20** bypass ratio is greater than about six (6), with an example embodiment being greater than about ten (10), the geared architecture **48** is an epicyclic gear train, such as a planetary gear system or other gear system, with a gear reduction ratio of greater than about 2.3 and the low pressure turbine **46** has a pressure ratio that is greater than about five. In one disclosed embodiment, the gas turbine engine **20** bypass ratio is greater than about ten (10:1), the fan diameter is significantly larger than that of the low pressure compressor **44**, and the low pressure turbine **46** has a pressure ratio that is greater than about five 5:1. Low pressure turbine **46** pressure ratio is pressure measured prior to inlet of low pressure turbine **46** as related to the pressure at the outlet of the low pressure turbine **46** prior to an exhaust nozzle. The geared architecture **48** may be an epicycle gear train, such as a planetary gear system or other gear system, with a gear reduction ratio of greater than about 2.3:1. It should be understood, however, that the above parameters are only exemplary of one embodiment of a geared architecture engine and that the present disclosure is applicable to other gas turbine engines including direct drive turbofans.

(18) A significant amount of thrust is provided by the bypass flow B due to the high bypass ratio. The fan section **22** of the gas turbine engine **20** is designed for a particular flight condition—typically cruise at about 0.8 Mach and about 35,000 feet (10,688 meters). The flight condition of

0.8 Mach and 35,000 ft (10,688 meters), with the engine at its best fuel consumption—also known as “bucket cruise Thrust Specific Fuel Consumption (‘TSFC’)”—is the industry standard parameter of lbm of fuel being burned divided by lbf of thrust the engine produces at that minimum point. “Low fan pressure ratio” is the pressure ratio across the fan blade alone, without a Fan Exit Guide Vane (“FEGV”) system. The low fan pressure ratio as disclosed herein according to one non-limiting embodiment is less than about 1.45. “Low corrected fan tip speed” is the actual fan tip speed in ft/sec divided by an industry standard temperature correction of  $[(T_{\text{am}} \text{ } ^\circ \text{R}) / (518.7^\circ \text{R})]^{.5}$ . The “Low corrected fan tip speed” as disclosed herein according to one non-limiting embodiment is less than about 1150 ft/second (350.5 m/sec).

(19) As will be described below, a fan blade is provided for use with the gas turbine engine **20** of FIG. **1** as well as other similar assemblies. Fan blades are typically made out of a variety of materials, including metal or fiber composites. When fiber composites are used fan blades, the fan blades need to have increased thickness relative to a metal blade, which tends to result in a decrease in fan efficiency. To make a fan blade out of fiber composite materials that also has a high level of aerodynamic efficiency requires that the thickness, chord and camber distributions of the fan blade all be tailored to enable the fan blade to withstand bird impacts while still delivering world class performance over full operating ranges.

(20) Thus, fan blade design is provided for, but not limited to, an 81” fan blade assembly with 16 blades, an average span of 27 inches per fan blade, a fan duct pressure ratio of 1.38 and a tip speed of 1050 ft/sec at cruise conditions, which is made out of carbon composite material or other similar materials. This new fan blade design incorporates thickness and camber distributions that improve bird impact capability while minimizing aerodynamic performance debit. The fan blade is an airfoil that combines both a high maximum thickness distribution with a forward location of maximum thickness with increased thickness in the forward portion of the airfoil in order to reduce stresses in the front portion due to the bird impact. Equally important, the airfoil incorporates a highly tailored camber distribution over the whole airfoil, optimized in both the core and duct regions, in order to maximize flow capacities and the efficiency of the fan blade over entire operating envelopes. This is particularly relevant in the region of span that affects the flow going into the core stream, which is dependent on the bypass ratio for a particular application (in an exemplary case, the span affected is from 0% to approximately 18% span).

(21) With continued reference to FIG. **1** and with additional reference to FIGS. **2** and **3** and FIG. **4**, a fan blade **201** is provided. FIGS. **2** and **3** are isometric and side views of a portion of the fan blade **201** that is configured to be disposed within a high-temperature and high-pressure flow path of, for example, a gas turbine engine, such as the gas turbine engine **20** of FIG. **1**. FIG. **4** is an illustration of a pair of adjacent airfoils showing a pre-compression region, a passage throat and a flat suction-surface in the post-throat diffuser region.

(22) As shown in FIGS. **2** and **3**, the fan blade **201** includes a body **202** with an airfoil shaped section (hereinafter referred to as the “airfoil section”) **210**. The airfoil section **210** has a root **211**, a tip **212**, leading and trailing edges **213** and **214** that extend between the root **211** and the tip **212** and pressure and suction surfaces **215** and **216** (see FIG. **4**).

(23) As shown in FIG. **4**, the fan blade **201** can be provided as a first fan blade **201.sub.1** and a second fan blade **201.sub.2** neighboring the first fan blade **201.sub.1** with a throat region **220** interposed between the suction surface **216** of the first fan blade **201.sub.1** and the pressure surface **215** of the second fan blade **201.sub.2**. The throat region **220** includes a passage throat **221** that is located at a minimum distance between the pressure surface **215** of the second fan blade **201.sub.2** and the suction surface **216** of the first fan blade **201.sub.1**. In accordance with embodiments, the first and second fan blades **201.sub.1** and **201.sub.2** are configured such that a pre-compression region **230** is defined in the throat region **220** ahead of the passage throat **221** and each of the first and second fan blades **201.sub.1** and **201.sub.2** includes a mean camber line MCL defining a flattened suction surface. In addition, each of the first and second fan blades **201.sub.1** and

**201.sub.2** includes an airfoil section that is defined on an axisymmetric surface with a symmetry axis of the surface aligned with a gas-path center-line axis and each of the first and second fan blades **201.sub.1** and **201.sub.2** has a symmetric thickness distribution with a single thickness maximum. This configuration maximizes flow-capacities and minimizes losses of the airfoil sections, while simultaneously maximizing a length of the passage 'diffuser' region between the passage throat **221** and the airfoil trailing-edges **213**.

(24) FIG. 5 is a graphical plot of a maximum thickness versus chord distribution for a fan blade, such as the fan blade **201** of FIGS. 2 and 3. The graphical plot shows that the maximum thickness versus chord distribution is relatively reduced between 0 and 50% span and particularly in the region between 20% and 50% span. FIG. 6A is planar view of the leading edge (LE) of an airfoil (i.e., of fan blade **201**) along the X-R plane and FIG. 6B is a plot of the angle of the LE on the X-R plane and illustrates that LE angle distribution in the X-R plane is correlated to bird impact capability of the blade (i.e., the fan blade **201**). FIG. 7A is a planar view of the LE of the airfoil (i.e., of fan blade **201**) along the Y-R plane and illustrates how angles that are plotted in FIG. 7B are calculated. FIG. 7B is a plot of the angle of the LE on the Y-R plane and illustrates that the LE angle distribution in the Y-R plane is correlated to the bird impact capability of the blade (i.e., the fan blade **201**). FIG. 8 is a plot of a distribution of airfoil thickness at 10% chord divided by the total chord (i.e., for fan blade **201**). FIG. 8 illustrates that a thickness in the front portion of the blade is relatively large particularly in the 0% to 90% span.

(25) With reference to FIGS. 5-8, the airfoil section **210** exhibits a range of thickness-over-chord (T/B) values of approximately 0.076 at about 20% span to approximately 0.05 at about 50% span, an average location of max thickness (LMT) of approximately 0.389 at about 0-20% span and an average of leading edge (LE) thickness at 10% chord-over-total chord of approximately 0.044 at 0-20% span (values listed herein are calculated for values interrogated at 5% increments along a span of the airfoil **201**; this regime will continue throughout the following description). More specifically, in about a 0%-20% span, the airfoil section **210** exhibits an average T/B of approximately 0.088, in about a 20%-50% span, the airfoil section **210** exhibits an average T/B of approximately 0.060, in about a 50%-90% span, the airfoil section **210** exhibits an average T/B of approximately 0.040 and in about a 90% to 100% span, the airfoil section **210** exhibits an average T/B of approximately 0.029. Further, in about a 0%-20% span, the airfoil section **210** exhibits an average LMT of approximately 0.389, in about a 20%-50% span, the airfoil section **210** exhibits an average LMT of approximately 0.422, in about a 50%-90% span, the airfoil section **210** exhibits an average LMT of approximately 0.501, and in about a 90% to 100% span, the airfoil section **210** exhibits an average LMT of approximately 0.556. Also, in about a 0%-20% span, the airfoil section **210** exhibits an average LE thickness of approximately 0.044, in about a 20%-50% span, the airfoil section **210** exhibits an average LE of approximately 0.029, in about a 50%-90% span, the airfoil section **210** exhibits an average LE of approximately 0.019, and in about a 90% to 100% span, the airfoil section **210** exhibits an average LE of approximately 0.013.

(26) In accordance with further embodiments, an angle of the LE, as assessed between adjacent sections stacked radially upon each other, affects how the incoming bird is impacted (and, i.e., sliced) by the fan blade **201**. For this fan blade **201**, a first region of the airfoil section **210**, which starts at about 35% span and extends to 75% span, but is not limited to these locations, is characterized in that the radial LE angle on a plane defined by X and R has increasing value to about 75% span (see FIG. 6A) with an inflection in the radial LE angle distribution at about 75%, but not limited to that location, above which there is a second region in which the radial LE angle decreases relative to the angle at about 75% span (see FIG. 6B). Here, a positive angle is defined by two radially adjacent sections, and the upper section is axially aft of the lower section (see FIG. 6A). Additionally or alternatively, for the radial LE angle on a plane defined by Y and R, a region of the airfoil section **210** that starts at about 35% span has an increasing radial angle to about 80% span (see FIG. 7A) with an inflection in the radial LE angle distribution at about 80%, above which



there is a region in which the radial LE angle decreases relative to the angle at about 80% span (see FIG. 7B). Here, a negative angle is defined by two radially adjacent sections, and the upper section is farther into the direction of rotation relative to the lower section (see FIG. 6A).

(27) The radial LE angles affect bird impact capability because the non-uniform radial LE shape profile allows a varying amount of blade material to cut into the bird, with varying amounts of energy transferred from the bird to the blade. Also, the force at the impact location is distributed in three coordinate directions due to the angled LE shape profile, and dissipates the energy into more advantageous areas of the airfoil.

(28) With reference to FIG. 8, the airfoil section **210** also exhibits a thickness-to-chord value at a 10% chord location at about 0% LE span of approximately 0.0585, which tapers to approximately 0.0396 at 11% LE span.

(29) In addition, the airfoil section **210** exhibits a chord distribution that has an inflection point between about 50% and about 70% span, the inflection point having a magnitude approximately 1.45-1.55 times a magnitude of the chord at 0% span, and approximately 1-1.1 times a magnitude of the chord at 100% span.

(30) FIG. 9A is a plot of an airfoil (i.e., of fan blade **201**) cross section in a transonic region of the airfoil and illustrates droop and flatness of the suction surface over the front portion of the airfoil as well as nearby a maximum thickness location on the airfoil the suction side surface remains flat but the pressure side surface deflects outwards as it accommodates increased thickness. FIG. 9B are plots of the metal angle distribution of the suction side, meanline and pressure side of the airfoil shown in FIG. 9A and illustrate a bump on the suction-side angle distribution near 40% chord of the airfoil as well as the nearly linear suction-side angle distribution in the aft portion of the airfoil. In addition, the thickness distribution over the chord of this airfoil section is also shown in the lower right plot, which shows that the airfoil has considerably increased thickness particularly in the front portion of the airfoil and at the max thickness location. FIG. 9C illustrates a corresponding predicted isentropic Mach number distribution over the surface of the airfoil section of FIG. 9A.

(31) With reference to FIGS. 9A, 9B and 9C and in accordance with embodiments, the airfoil section **210** can be tailored, especially in a sensitive transonic region of the airfoil span, to compensate for required increases in airfoil thickness for impact-damage resistance while minimizing negative effects on aerodynamics. Such a tailored airfoil section **210** is illustrated in FIGS. 9A and 9B with a resultant surface isentropic mach-number loading plot illustrated in FIG. 9C, computed at the airfoil aerodynamic design point.

(32) As shown in FIG. 9B in particular, the airfoil suction surface exhibits droop over the first 5% of airfoil chord, after which the suction surface metal-angle distribution is kept approximately flat, up to approximately -5% of the chord location where the adjacent airfoil 'covers' the airfoil. Following the flat suction-surface region, the suction-surface metal-angle distribution exhibits a subtle 'bump' between -5% & +5% of the airfoil covered-passage starting position. Following the precompression region, the suction-surface metal-angle distribution is approximately linear, except at the front and endpoints of this region where the linear section blends into the precompression and trailing-edge locations respectively. With the desired suction-surface now achieved, the thickness distribution is applied such that the increase in section maximum-thickness is evidenced on the pressure-surface only. The chordwise location of section maximum-thickness is positioned as optimally as possible to balance the favorable effect of moving the chordwise location of section maximum-thickness closer to the leading-edge for impact-damage resistance, versus the unfavorable effect moving the chordwise location of section maximum-thickness forward has on airfoil operating range, loss and shock-boundary-layer interaction (if a shock is present). The combination of the features described herein creates a pressure-surface metal-angle distribution that displays a pronounced concave distribution between the position of maximum-thickness and the trailing-edge as shown in FIG. 9B.

(33) As additional embodiments, the airfoil section **210** is formed to create a slight pre-compression region before the airfoil passage shockwave as shown in the circled feature in the Mach-number plot of FIG. **9C**. Here, pre-compression acts to slightly reduce the pre-shock Mach number thus reducing airfoil loss and making shock-boundary-layer-interaction less likely. The combination of the features described herein result in a concavity that transfers loading (difference in suction versus pressure surface Mach-number, or equivalently static-pressure) from the suction-surface to the pressure-surface, as indicated by the two downward pointing arrows in FIG. **9C** at approximately 45% and 80% of airfoil chord respectively.

(34) The transfer of the airfoil loading in this manner allows the airfoil to accommodate increased airfoil thickness requirements, without compromising sensitive suction-surface aerodynamics. Transferring the loading to the pressure-surface aft region also helps to reduce airfoil profile loss, as the equivalent loading is occurring at a lower overall passage Mach number (profile losses scale with the cube of the boundary-layer edge velocity, so small reductions in the boundary-layer-edge velocity can yield useful gains in loss reduction).

(35) The resultant effect of these geometric features in combination is an efficient airfoil section **210** and a resulting blade passage that exhibits low loss and high stalling static-pressure-rise capability. It also suppresses the formation of passage loss cores, when the trailing-edge mean camber line metal-angle is below ~7 degrees or greater than ~20 degrees.

(36) Benefits of the features described herein are the provision of a fan blade in which a thickness distribution and a tailored camber distribution are incorporated into the fan blade design allowing for the fan blade to be made out of a composite material and remain competitive for aerodynamic performance. The thickness distribution reduces tensile and compressive stresses experienced during a bird impact event, while the camber distribution enables world-class performance by minimizing the pressure and shock losses, therefore maximizing efficiency.

(37) The term “about” is intended to include the degree of error associated with measurement of the particular quantity based upon the equipment available at the time of filing the application.

(38) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the present disclosure. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, element components, and/or groups thereof.

(39) While the present disclosure has been described with reference to an exemplary embodiment or embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the present disclosure. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the present disclosure without departing from the essential scope thereof. Therefore, it is intended that the present disclosure not be limited to the particular embodiment disclosed as the best mode contemplated for carrying out this present disclosure, but that the present disclosure will include all embodiments falling within the scope of the claims.

## Claims

1. A gas turbine engine, comprising: a first fan blade comprising a suction surface; a second fan blade comprising a pressure surface and neighboring the first fan blade; and a throat region interposed between the suction surface of the first fan blade and the pressure surface of the second fan blade, the throat region comprising a passage throat located at a minimum distance between the pressure surface of the second fan blade and the suction surface of the first fan blade, the passage throat extending from a point of maximum thickness of the second fan blade and being normally

angled with respect to the pressure surface of the first fan blade, and the first and second fan blades being configured such that a pre-compression region is defined in the throat region ahead of the passage throat and each of the first and second fan blades comprising a mean camber line defining a flattened suction surface.

2. The gas turbine engine according to claim 1, wherein each of the first and second fan blades comprises an airfoil section that is defined on an axisymmetric surface.

3. The gas turbine engine according to claim 1, wherein each of the first and second fan blades has a single thickness maximum.

4. A gas turbine engine, comprising: an airfoil; and an adjacent airfoil, which is adjacent to the airfoil and which at least partially overlaps the airfoil in a circumferential dimension, wherein the airfoil comprises: a suction surface exhibiting droop over a first 5% of an airfoil chord, after which a suction surface metal-angle distribution decreases flatly along a flat suction surface region to where the adjacent airfoil at least partially overlaps the airfoil in the circumferential dimension, wherein: following the flat suction surface region, the suction surface metal-angle distribution exhibits a bump of an airfoil covered-passage starting position, and following a precompression region, the suction surface metal-angle distribution is linear along a linear section, except at front and end points of the linear section, where the linear section blends into precompression and trailing-edge locations, respectively.

5. The gas turbine engine according to claim 4, wherein an increase in a section maximum-thickness is on a pressure surface only of the airfoil.

---